

General Assignment (3): Artificial Neural Network (ANN) for Medical Diagnosis

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Introduction

This assignment aims to develop and evaluate an Artificial Neural Network (ANN) for medical diagnosis, specifically predicting diabetes. The process involves comprehensive data understanding, preprocessing, model design, training, hyperparameter tuning, and evaluation. The goal is to build a reliable model capable of assisting in the early identification of diabetes.

Task 1: Utilise Libraries / Dataset

Activity 1: Import all necessary

```
# Task 1, Activity 1: Import all necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split as tts
from sklearn.preprocessing import StandardScaler as ssc
from sklearn.metrics import classification_report as crp
from sklearn.metrics import confusion_matrix as cmt

import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
import os
print("Libraries imported successfully!")
```

Libraries imported successfully!

Activity 2: Load the diabetes dataset and display the first few rows

Load the data set from google drive

```
# Task 1 - Activity 2 - Load the diabetes dataset and display the first few rows
import pandas as pd
from google.colab import drive
drive.mount('/content/drive')

# Only access a specific project folder
project_path = '/content/drive/My Drive/Colab Notebooks/NUS_GenAI_Lab/lab_data'

enhanced_diabetes_dataset = f'{project_path}/enhanced_diabetes_dataset.csv'

try:
    # Read the data from CSV file
    df = pd.read_csv(enhanced_diabetes_dataset)
    print("Successfully loaded data from enhanced_diabetes_dataset.csv")

except FileNotFoundError:
    print("enhanced_diabetes_dataset.csv file not found.")

# Display the first few rows of the dataset
print("\nFirst 5 rows of the enhanced diabetes dataset dataset:")
print(df.head())
print("record")
print(df['Diabetes'].value_counts())
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
Successfully loaded data from enhanced_diabetes_dataset.csv

First 5 rows of the enhanced diabetes dataset dataset:

	Age	Gender	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	\
0	52	1	1	124	95.0	20.0	0	
1	43	1	2	144	86.0	23.0	3	
2	55	1	4	141	113.0	18.0	297	
3	68	1	3	125	110.0	32.0	210	
4	41	0	0	177	92.0	14.0	189	

	BMI	DiabetesPedigreeFunction	HbA1c	FastingBS	Triglycerides	HDL	\
0	36.0	0.078	4.7	87	142.0	37.0	
1	32.9	0.118	6.3	135	162.0	89.0	
2	43.4	0.139	5.4	113	180.0	45.0	
3	22.7	0.197	5.2	93	262.0	56.0	
4	38.1	0.078	10.5	126	332.0	66.0	

Activity 3: Show basic information and statistics about the dataset

Understand the features of the dataset, including their relevance to medical diagnosis, and identify the target variable.

Understanding: The dataset contains various medical and demographic features relevant to diabetes prediction. These include:

- **Age:** Patient's age in years.
- **Gender:** Patient's gender (0=Male, 1=Female).
- **Pregnancies:** Number of times pregnant.
- **Glucose:** Plasma glucose concentration (mg/DL).
- **BloodPressure:** Diastolic blood pressure (mm Hg).
- **SkinThickness:** Triceps skin fold thickness (mm).
- **Insulin:** 2-Hour serum insulin (mu U/ml).
- **BMI:** Body mass index (weight in kg/(height in m)^2).
- **DiabetesPedigreeFunction:** A function that scores the likelihood of diabetes based on family history.
- **HbA1c:** Hemoglobin A1c level.
- **FastingBS:** Fasting blood sugar.
- **Triglycerides:** Triglyceride levels.
- **HDL:** High-density lipoprotein cholesterol levels.
- **Smoking:** Smoking status.
- **PhysicalActivity:** Physical activity level.
- **Prediabetes:** Indicator for prediabetes

- **Diabetes:** 0=No diabetes 1= diabetes.

```

Dataset information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2000 entries, 0 to 1999
Data columns (total 17 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Age                                    2000 non-null   int64
1   Gender                                2000 non-null   int64
2   Pregnancies                           2000 non-null   int64
3   Glucose                                2000 non-null   int64
4   BloodPressure                          1902 non-null   float64
5   SkinThickness                          1914 non-null   float64
6   Insulin                                 2000 non-null   int64
7   BMI                                    2000 non-null   float64
8   DiabetesPedigreeFunction                2000 non-null   float64
9   HbA1c                                  2000 non-null   float64
10  FastingBS                              2000 non-null   int64
11  Triglycerides                           1909 non-null   float64
12  HDL                                     1904 non-null   float64
13  Smoking                                 2000 non-null   int64
14  PhysicalActivity                        1899 non-null   float64
15  Prediabetes                             2000 non-null   int64
16  Diabetes                                2000 non-null   int64

```

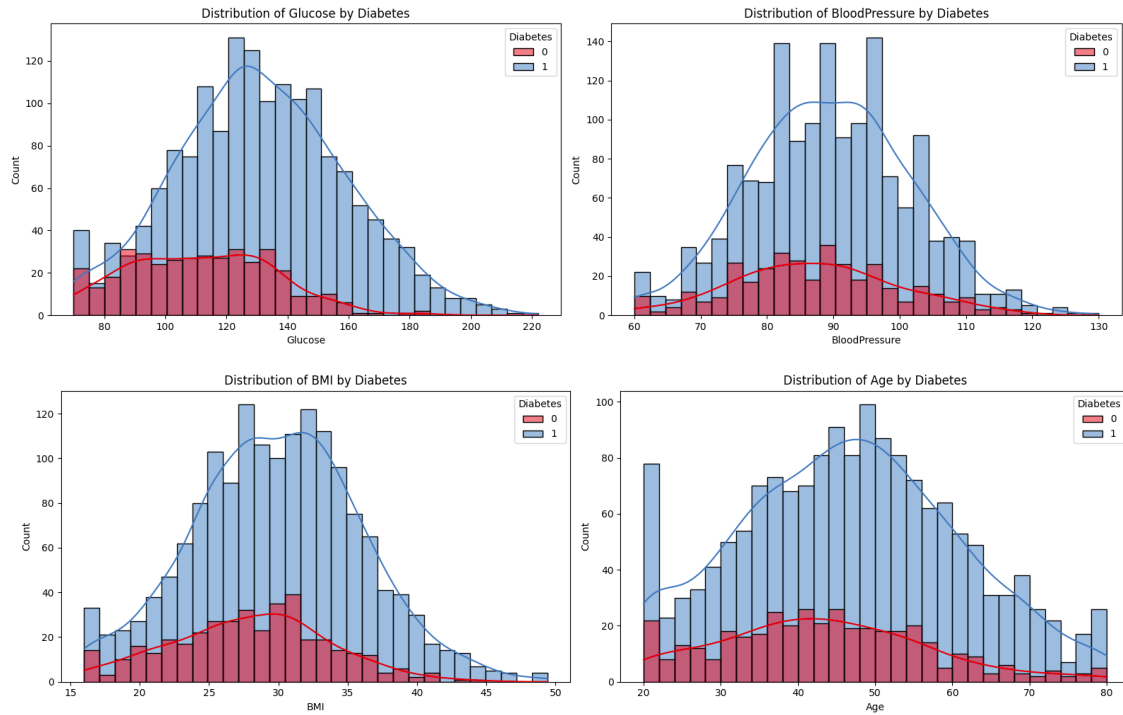
Summary statistics:

	Age	Gender	Pregnancies	Glucose	BloodPressure \
count	2000.000000	2000.000000	2000.000000	2000.000000	1902.000000
mean	45.858500	0.614000	1.557000	127.156000	89.005258
std	14.155164	0.486952	1.758208	28.018051	12.119690
min	20.000000	0.000000	0.000000	70.000000	60.000000
25%	36.000000	0.000000	0.000000	107.000000	81.000000
50%	46.000000	1.000000	1.000000	126.000000	89.000000
75%	55.000000	1.000000	3.000000	146.000000	97.000000
max	80.000000	1.000000	10.000000	222.000000	130.000000

Task 2: Generate at least three EDA visualisations

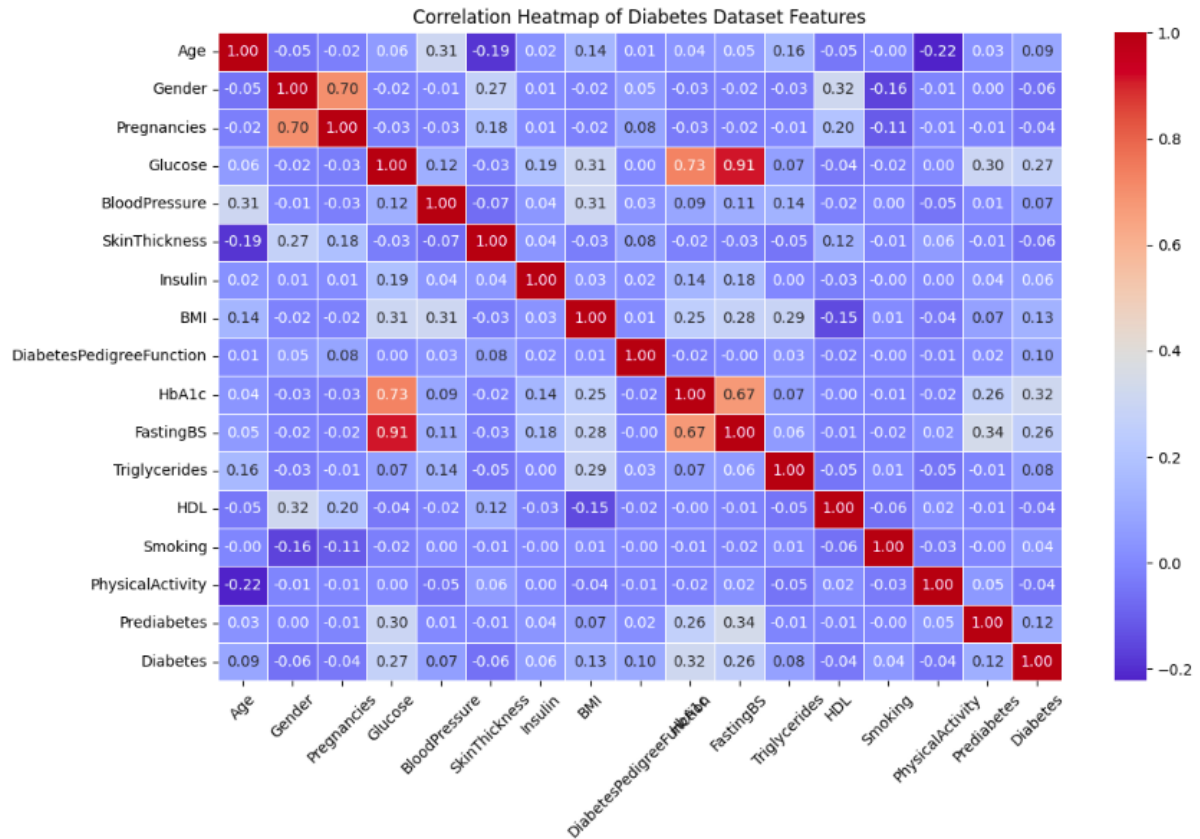
Activitiy 1 – Create distribution plots comparing features between diabetic and non-diabetic patients

Below diagram displays the distribution of Glucose, Blood Pressure, BMI, and Age, separated by Diabetes status (0=non-diabetic, 1=diabetic). It shows visual comparison of how these key features differ between the two patient groups, aiding in understanding their relationship with diabetes.



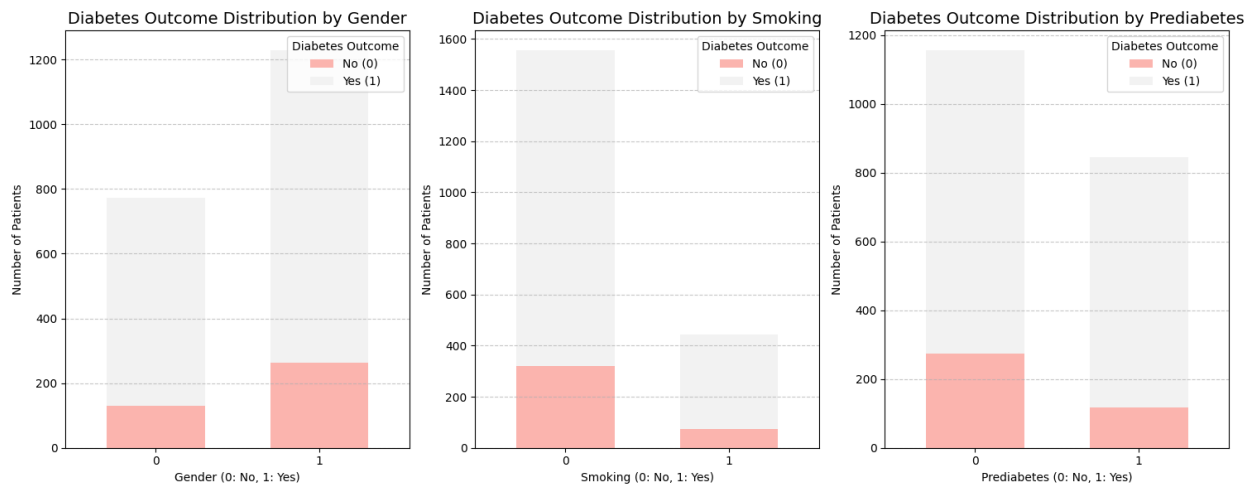
Activitiy 2 - Develop correlation heatmap to identify relationships between features

This heatmap visualizes the correlation matrix of diabetes dataset features. Red indicates positive correlation, blue indicates negative correlation. It highlights strong relationships, between Glucose and FastingBS with Diabetes.



Activitiy 3 - Visualise feature importance using appropriate techniques

These count plots illustrate the absolute counts of patients with and without diabetes across categories of binary features like Gender, Smoking, and Prediabetes.



Task 3: Analyse Data Quality

Activity 1: Check for missing or zero values

SkinThickness 86 record value is zero.

```
Number of NaN (Not a Number) missing values per column:
Age 0
Gender 0
Pregnancies 0
Glucose 0
BloodPressure 98
SkinThickness 86
Insulin 0
BMI 0
DiabetesPedigreeFunction 0
HbA1c 0
FastingBS 0
Triglycerides 91
HDL 96
Smoking 0
PhysicalActivity 101
Prediabetes 0
Diabetes 0
```

Activity 2: Identify and handle outliers

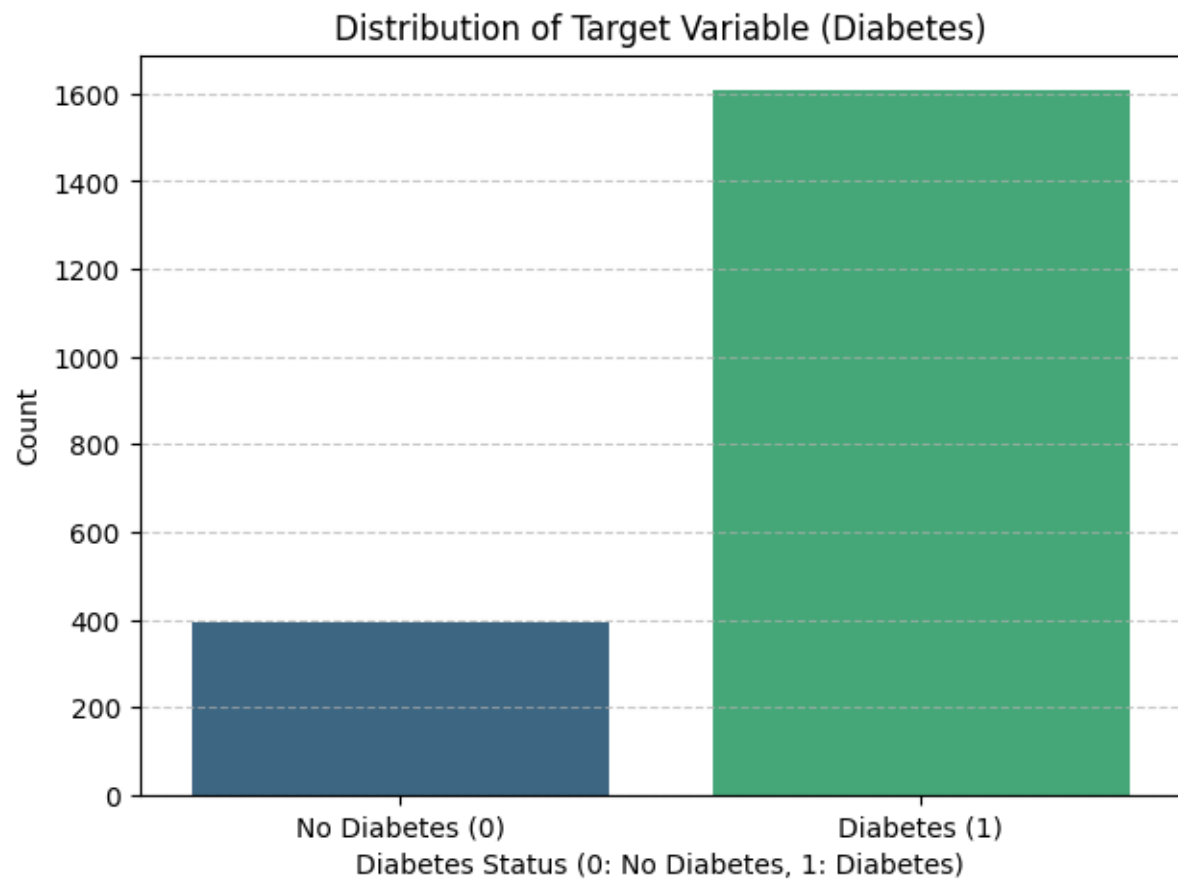
Outliers significantly affect model training. The method used to identify outliers is Interquartile Range (IQR) method.

```
Total outliers capped per feature:
- Pregnancies: 6
- Glucose: 6
- BloodPressure: 6
- SkinThickness: 9
- Insulin: 1
- BMI: 8
- DiabetesPedigreeFunction: 106
- HbA1c: 5
- FastingBS: 8
- Triglycerides: 7
- HDL: 4
- PhysicalActivity: 54
```

Task 4: Prepare Features and Target Variable

Activity 1: Split data into features (X) and target (y)

Bar graph showing the distribution of diabetes cases generated



Activity 2: Normalise/standardise features

Splitting the dataset into training validation and test set.

--- Normalising/Standardising Features using StandardScaler ---

X_train_scaled shape: (1100, 16)

X_val_scaled shape: (300, 16)

X_test_scaled shape: (600, 16)

First 5 rows of X_train_scaled (example):

	Age	Gender	Pregnancies	Glucose	BloodPressure	SkinThickness	\
0	-0.695838	0.813407	-0.318460	-0.152845	0.724994	0.591398	
1	-1.330049	0.813407	0.806118	-0.047313	-0.512165	1.160993	
2	-0.413967	-1.229396	-0.880749	0.585877	0.312608	-0.661711	
3	-0.836774	-1.229396	-0.880749	0.796941	-0.347210	0.591398	
4	-0.695838	0.813407	0.806118	-0.223200	-1.007028	0.249641	

	Insulin	BMI	DiabetesPedigreeFunction	HbA1c	FastingBS	\
0	-0.303502	0.096537		-0.816642	-0.618474	1.123828
1	-0.948137	-0.393347		0.607447	0.358628	0.263795
2	1.329573	0.766045		-0.916198	0.875917	0.521805
3	-0.664497	1.451883		0.092351	1.508160	0.951822
4	1.458500	-2.189589		-0.007205	-1.135764	0.263795

	Triglycerides	HDL	Smoking	PhysicalActivity	Prediabetes
0	-0.954575	1.166519	1.825023	0.957085	1.162079
1	-0.520760	-1.147062	-0.547938	NaN	1.162079
2	1.792923	-1.816782	-0.547938	-1.074766	1.162079
3	0.443275	0.192380	1.825023	0.535380	1.162079
4	-1.376340	0.070612	-0.547938	-0.423040	1.162079

Task 5: Handling Class Imbalance and Dataset Splitting

Activity 1: Handle class imbalance (e.g., using SMOTE)

Divide the preprocessed dataset into training, validation, and test sets for model development and unbiased evaluation.

Class distribution in original y_train:

Diabetes

1 1286

0 314

Name: count, dtype: int64

Diabetes

1 0.80375

0 0.19625

Name: proportion, dtype: float64

Activity 2: Split data into training (70%), validation (15%), and test (15%) sets

Data successfully split into training, validation, and test sets.

--- Splitting Data into Training, Validation, and Test Sets (70%-15%-15%) ---

Original dataset shape (X): (2000, 16)

Original dataset shape (y): (2000,)

--- Resulting Set Shapes ---

Training set (70%): X_train: (1400, 16), y_train: (1400,)

Validation set (15%): X_val: (300, 16), y_val: (300,)

Test set (15%): X_test: (300, 16), y_test: (300,)

Task 6: Design an Appropriate DCNN-based Architecture

Activity 1: Determine appropriate number of layers and neurons

This image displays a Keras model summary for an Artificial Neural Network. It shows three Dense layers: the first with 64 neurons, the second with 32, and the output layer with 1 neuron. It has a total of 3,201 trainable parameters, indicating the model's complexity and learned weights.

Layer (type)	Output Shape	Param #
dense_6 (Dense)	(None, 64)	1,088
dense_7 (Dense)	(None, 32)	2,080
dense_8 (Dense)	(None, 1)	33

Total params: 3,201 (12.50 KB)
Trainable params: 3,201 (12.50 KB)
Non-trainable params: 0 (0.00 B)

Task 7: Configure the Training Process

Regularisation technique implemented to prevent overlifting.

Task 8: Train the model with Appropriate Batch Size and Number of Epochs

Training the Artificial Neural Network model using the specified batch size and number of epochs.

```
Epoch 1/100  
44/44 ----- 4s 27ms/step - accuracy: 0.1897 - auc: 0.0041 - loss: nan - precision: 0.3333 - recall: 0.0075 - val_accuracy: 0.1967 - val_auc: 0.0000e+00 - val_loss: nan - val_  
Epoch 2/100  
44/44 ----- 0s 7ms/step - accuracy: 0.1924 - auc: 0.0000e+00 - loss: nan - precision: 0.0000e+00 - recall: 0.0000e+00 - val_accuracy: 0.1967 - val_auc: 0.0000e+00 - val_loss:  
Epoch 3/100  
44/44 ----- 0s 9ms/step - accuracy: 0.2170 - auc: 0.0000e+00 - loss: nan - precision: 0.0000e+00 - recall: 0.0000e+00 - val_accuracy: 0.1967 - val_auc: 0.0000e+00 - val_loss:  
Epoch 4/100  
44/44 ----- 0s 4ms/step - accuracy: 0.1934 - auc: 0.0000e+00 - loss: nan - precision: 0.0000e+00 - recall: 0.0000e+00 - val_accuracy: 0.1967 - val_auc: 0.0000e+00 - val_loss:  
Epoch 5/100  
44/44 ----- 0s 4ms/step - accuracy: 0.1979 - auc: 0.0000e+00 - loss: nan - precision: 0.0000e+00 - recall: 0.0000e+00 - val_accuracy: 0.1967 - val_auc: 0.0000e+00 - val_loss:
```

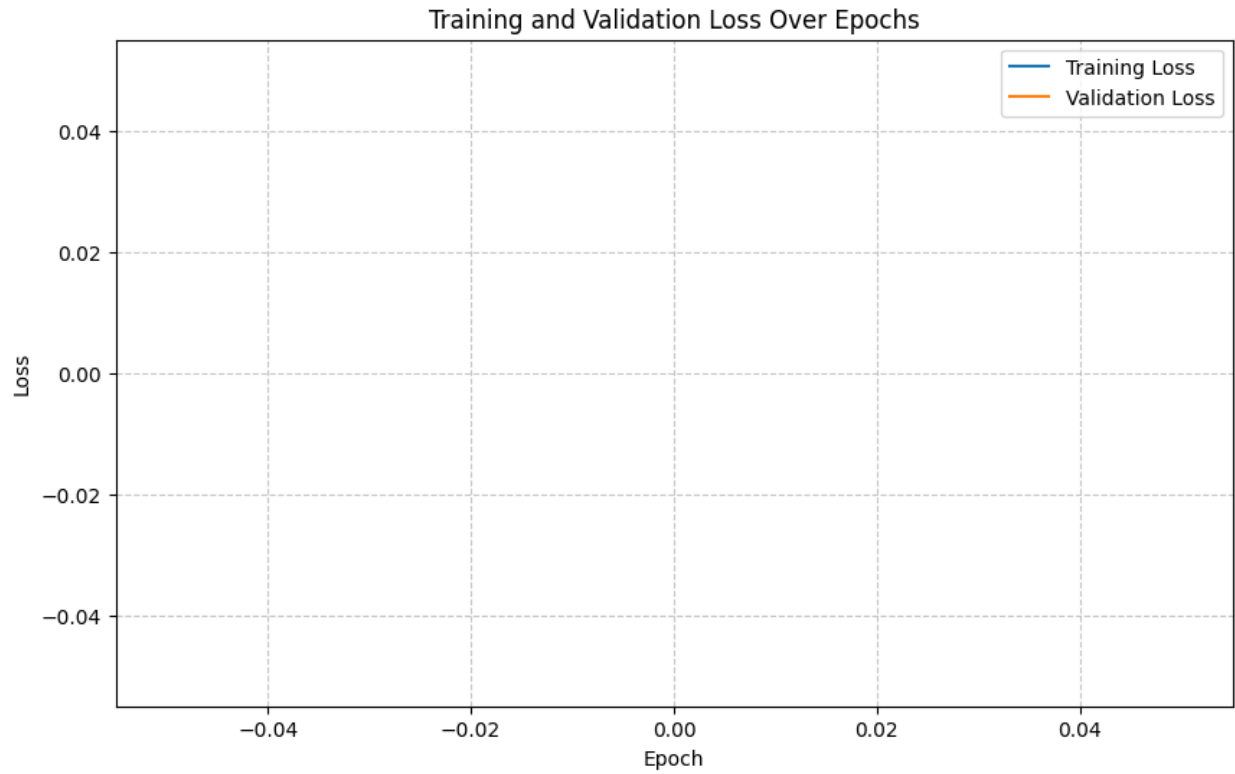
Task 9: Implement Callbacks

Early stopping to prevent overfitting, mModel checkpointing to save the best model and learning rate scheduling if appropriate implemented.

Task 10: Visualise the Training Process

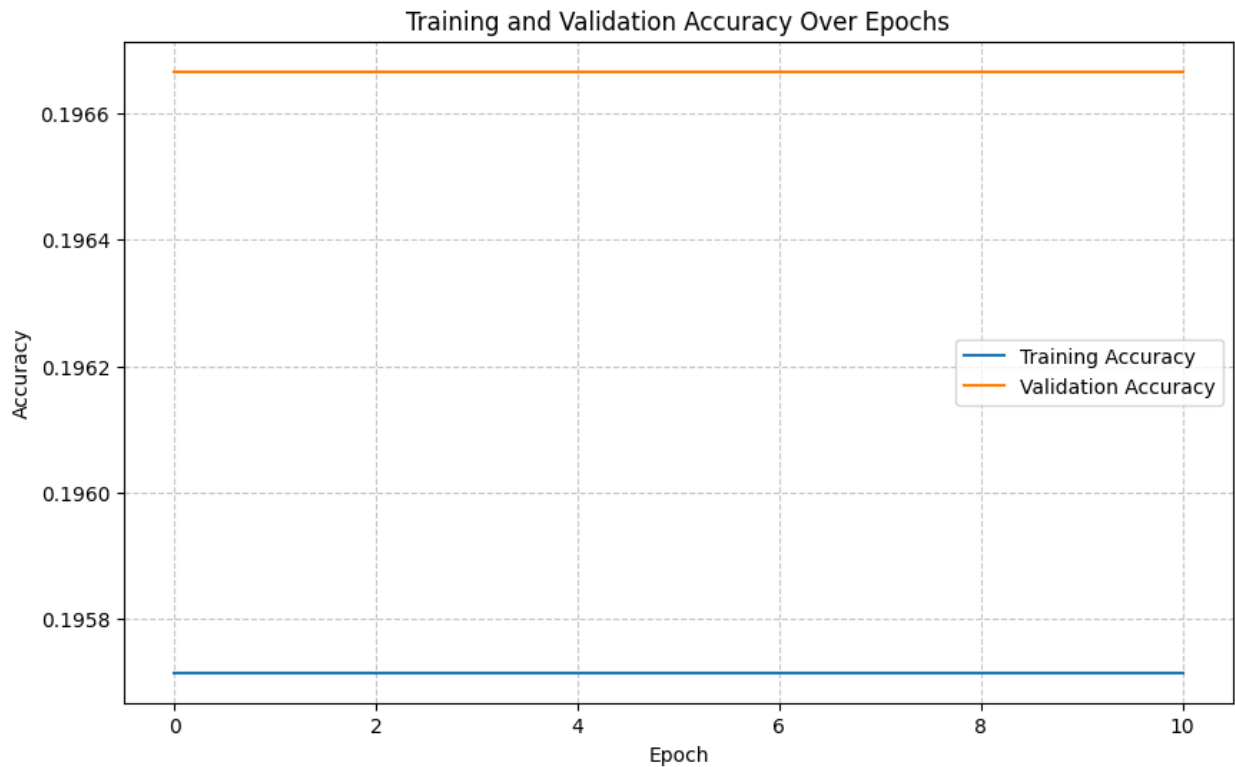
Activity 1: Plot training and validation loss

Plot suppose to show the training loss and validation loss over epochs to visualize convergence and potential issues. Some how no data shown.



Activity 2: Plot training and validation loss

Training and validation accuracy graph



Task 12: Evaluate the Model on the Test Dataset

The confusion matrix shows the model predicted all cases as "no diabetes" (0). It failed to identify any true positives (1), indicating poor performance with 0% sensitivity and high bias towards the majority class.

